

# Find non-benchmark percentages

Use the scaffold, show each step and ask for an adult check when marked.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Read the model: Find 35% of 480. Copy or complete the first step.

2. Find 17% of 650.

3. A £240 coat is reduced by 15%. What is the sale price?

4. Miro says: Miro subtracts the percentage number rather than the percentage amount. Is this correct? Explain.

# Teacher answers and checking prompts

1. Read the model: Find 35% of 480. Copy or complete the first step.

**Answer** 168.

2. Find 17% of 650.

**Answer** 110.5.

3. A £240 coat is reduced by 15%. What is the sale price?

**Answer** £204.

4. Miro says: Miro subtracts the percentage number rather than the percentage amount. Is this correct? Explain.

**Answer** Calculate the percentage of the original amount first, then apply that change in the context.

# Find non-benchmark percentages

Work independently. Show a complete method and check at least one answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Find 35% of 480.

6. Check this result using an inverse, estimate or known fact: A £240 coat is reduced by 15%. What is the sale price?

2. Find 17% of 650.

7. Prove or explain your reasoning: 84 is 35% of a number. What is the number?

3. A £240 coat is reduced by 15%. What is the sale price?

8. Identify and correct this misconception: Miro subtracts the percentage number rather than the percentage amount.

4. Solve this independently and show every stage: Find 35% of 480.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Find 17% of 650.

10. Write and solve a similar question to: A £240 coat is reduced by 15%. What is the sale price?

# Teacher answers and checking prompts

1. Find 35% of 480.  
**Answer** 168.
2. Find 17% of 650.  
**Answer** 110.5.
3. A £240 coat is reduced by 15%. What is the sale price?  
**Answer** £204.
4. Solve this independently and show every stage: Find 35% of 480.  
**Answer** 168.
5. Use a second method or representation for: Find 17% of 650.  
**Answer** Expected result: 110.5. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A £240 coat is reduced by 15%. What is the sale price?  
**Answer** Expected result: £204. A valid check must be shown.
7. Prove or explain your reasoning: 84 is 35% of a number. What is the number?  
**Answer** 240. The explanation must show why the method works.
8. Identify and correct this misconception: Miro subtracts the percentage number rather than the percentage amount.  
**Answer** Calculate the percentage of the original amount first, then apply that change in the context.
9. Write a complete sentence explaining how you checked one answer.  
**Answer** Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A £240 coat is reduced by 15%. What is the sale price?  
**Answer** Answers vary. The new question must use the same mathematical structure and include a correct solution.

# Find non-benchmark percentages

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Prove or explain your reasoning: 84 is 35% of a number. What is the number?

6. Create a new example that tests the same idea as: A £240 coat is reduced by 15%. What is the sale price?

2. Prove the answer to this question, rather than only calculating it: Find 35% of 480.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Find 17% of 650.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: £204.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro subtracts the percentage number rather than the percentage amount.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

# Teacher answers and checking prompts

1. Prove or explain your reasoning: 84 is 35% of a number. What is the number?  
**Answer** 240. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Find 35% of 480.  
**Answer** Expected result: 168. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Find 17% of 650.  
**Answer** Expected result: 110.5. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: £204.  
**Answer** One valid reconstruction is: A £240 coat is reduced by 15%. What is the sale price? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro subtracts the percentage number rather than the percentage amount.  
**Answer** Calculate the percentage of the original amount first, then apply that change in the context.
6. Create a new example that tests the same idea as: A £240 coat is reduced by 15%. What is the sale price?  
**Answer** Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.  
**Answer** Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.  
**Answer** Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.  
**Answer** Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.  
**Answer** Answers vary. The explanation must distinguish the mathematical structure from the changed value.